

Build Log

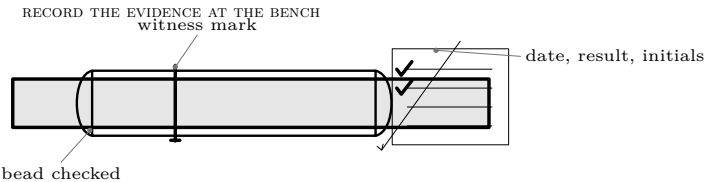
Print it, fill it in with a pen, keep it with the launcher

*One sheet, one build — the evidence that it
is sound*

This is the difference between a launcher you hope is fine and one you know is fine. Fill it in as you go, not afterwards from memory. Every blank on this page exists because somebody, somewhere, guessed instead of checking.

How to use this log

Read the governing build procedure and craft sheets before work. Then keep the following record packet together and complete each table at the named hold point. Do not reconstruct entries later. A blank, uncertainty, or failed check is a hold; the adult in charge decides whether the affected part is remade or retired.



Record sequence

1. Identify the governing drawing.
2. Name the adult with stop authority.
3. Copy every pressure-part rating before measuring; an absent or nonmatching rating places that part on HOLD.
4. Record each cut and edge inspection before joining.
5. Record each joint while it is fresh; any blank or failed observation places the affected assembly on HOLD.
6. Calculate the cure gate from the latest weld.
7. Close every fault by accept, remake, or retire.
8. Complete final visual proof only after cure.
9. Complete the session sheet before each later use.

RECORDS NEVER OVERRULE A DEFECT

Good paperwork cannot make a cracked, crazed, distorted, misaligned, unknown, or incompletely cured pressure part acceptable. Stop, mark the fault, isolate the part, and follow the remake record. Firing is never an inspection method.

Consolidated build record

The explanatory procedure belongs in the build guide. The pages that follow are the bench record: identity, measurements, joint evidence, faults, release, and session custody, in that order.

Builder _____
Adult in charge _____
Build started _____ **Finished** _____
Launcher ID _____ **Log page** _____ of _____

Before work — questions that must have answers

Question	Answer / evidence location
Which drawing revision governs this build?	_____
Are all pressure parts new, undamaged, and traceable?	_____
Which adult may stop the work without discussion?	_____
Where will rejected parts be marked and isolated?	_____

Materials — copy from the printing on the pipe

Part	Nominal	Schedule / rating	Marking says (copy it)
Chamber pipe	_____	_____	_____
Barrel pipe	_____	_____	_____
Reducer	_____	_____	_____
Cleanout	_____	_____	_____

STOP HERE IF ANY LINE DOES NOT SAY PRESSURE-RATED

Every pressure-bearing part must read **Schedule 40** with an **NSF-PW** rating and a pressure figure. If a part says DWV, cellular core, foam core, or carries no pressure rating at all, it does not go in this build. Mark it REJECT, isolate it from accepted stock, and obtain a traceable replacement before continuing.

Cut list — measured twice, from one reference end

Piece	Target	1st measure	2nd	Square? (4 positions)
Chamber	_____	_____	_____	_____
Barrel	_____	_____	_____	_____

GOVERNED BY CRAFT 1 AND CRAFT 2

Measurement record. Tool used _____ Tool condition checked _____ Reference end marked _____

If the two readings disagree, do not average them. Find the cause, mark the piece HOLD, and record the decision on the fault page.

Every end deburred and chamfered

End	Inside	Outside	Checked by (initials)
Chamber, breech end			
Chamber, reducer end			
Barrel, reducer end			
Barrel, muzzle			

GOVERNED BY CRAFT 3

First inspection hold point

Evidence reviewed	Pass / hold	Inspector, date, time
Ratings copied and legible on the actual parts		
Lengths independently re-read from the reference end		
Four-position squareness and every edge examined		

A BLANK IS A HOLD, NOT A PASS
The inspector asks: What did you observe? What did you measure? Where is it written? If an answer depends on memory, stop and make the observation again.

Joints — the page that matters most

For each joint: the date and time you welded it, whether primer was used, whether the squeeze-out bead was continuous all the way round with no gaps, and whether the witness mark stayed aligned.

Joint	Date & time	Primer	Bead all round	Witness mark aligned
Cleanout to chamber	_____	_____	_____	_____
Chamber to reducer	_____	_____	_____	_____
Reducer to barrel	_____	_____	_____	_____

GOVERNED BY CRAFT 4

Joint-by-joint inspection record

Joint	Dry fit / insertion mark	Observed defect?	Disposition / initials
Cleanout to chamber			
Chamber to reducer			
Reducer to barrel			

Intermediate question. Is the bead truly continuous, or merely hidden from view? Rotate the whole assembly once under good light. Record any gap, movement, misalignment, contamination, or uncertainty as a fault. Any such observation places the assembly on HOLD; reject and remake it rather than filling or “touching up” the joint.

THE CURE GATE — COMPUTE IT, DO NOT ESTIMATE IT

Latest weld finished at _____ (date and time).

Add 48 hours. **Earliest permitted first firing:** _____

Write that time here, and write it on the chamber in marker as well. Nobody fires this before then — not for a quick test, not because it feels solid, not because the weather is good today and bad on Sunday.

Igniter

Check	Result / initials
Holes round, no crazing, no cracks	_____
Electrode gap measured	_____ mm _____
Spark seen <i>and</i> heard, chamber open, before sealing	_____
Epoxy fillet continuous all round, both electrodes	_____
Epoxy fully cured before first pressure	_____

GOVERNED BY CRAFT 5

Igniter inspection note. Lighting used _____ Magnification used _____ Inspection date/time _____

Question raised during inspection: _____

Evidence that resolved it: _____

Fault, hold, and remake record

Use one row for every departure from the drawing, procedure, or expected appearance. “No fault found” is a result only after the inspection is written down. Rejected pressure parts are never repaired for launcher use.

ID	Observation / measurement	Immediate hold	Decision and reason	Verified by
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Permitted dispositions
Accept only when the drawing and every stated criterion are met. Remake means a new part or assembly, followed by every earlier check that the replacement affects. Retire means mark it permanently, isolate it, and make it unusable. There is no “use carefully,” “probably fine,” or performance-based exception.

Remake trace

Fault ID	Removed part marked / isolated	Replacement identity	Checks repeated; pages / initials
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Final dimensional and visual proof

Proof item	Observed / measured	Witness, date, time
Overall assembly matches the governing drawing		
Chamber and barrel identity markings remain legible		
All joint beads viewed through a full rotation		
No crack, craze, whitening, deformation, or loose part		
Cure-gate calculation independently checked		
All fault IDs closed by accept, remake, or retire		

FINAL RELEASE IS NOT A FIRING TEST
Do not use firing to prove that construction is sound. Release means only that the documented build and inspection criteria have been met. It does not waive the separate pre-firing inspection, site controls, or adult stop authority.

Pre-firing inspection — all ten, every session

Run the Craft 6 list. Initial each. Any single failure stops the session.

Check	Session 1	2	3
1 Cut faces square			
2 Edges deburred and chamfered			
3 Every weld bead continuous			
4 Cure time elapsed			
5 Penetrations sound, no crazing			
6 Chamber wall: no crazing or cracks			
7 Cleanout threads clean, seats fully			
8 Spark confirmed, chamber open			
9 Barrel bore clear (checked from breech)			
10 Pressure rating still legible on every part			
GOVERNED BY CRAFT 6			

Session record — observation, not optimization

Date	Adult in charge	Site / backstop	Inspection result	Unexpected observation / stop decision

Retirement. Any crazing, whitening, star crack or deformation of the chamber retires it permanently — cut it up so it cannot be reused and build a new one. Record the date and the reason here:

Final proof and custody

Statement	Name / initials / date
Builder: entries were made at the time of the work	
Inspector: evidence was checked, not reconstructed from memory	
Adult in charge: open faults and holds are understood	
Log stored with launcher at	

Why a form, and not just care
Memory is confident and wrong. “I’m fairly sure” is not evidence. A dated entry removes the argument before it begins and lets another person verify the same facts.